Appendix E Constructing PUMA-to-CZ Crosswalks

Appendix E.1 Overview

Commuting zones (CZs) are clusters of counties linked by commuting flows, originally developed by Tolbert and Sizer (1996) to delineate local labor markets that cover the entire United States. Two CZ vintages are currently available: CZ 1990, the long-standing definition built from 1990 commuting flows and popularized by Autor et al. (2019); Autor and Dorn (2013), and CZ 2020, a new definition recently released by the USDA Economic Research Service based on the methodology of Fowler (2024).

Public survey data such as the ACS usually do not directly identify commuting zones. PUMAs are the smallest geographic units identified in the ACS, and researchers must map PUMAs to CZs using crosswalks. The most widely used approach in the economics literature follows Autor and Dorn (2013) and Autor et al. (2019), which provides crosswalks from PUMAs to CZ 1990 with population-share allocation factors for PUMA vintages up to 2010.

A complication is that the Census Bureau redraws PUMA boundaries every ten years based on population information from the most recent decennial census. Under this redrawing, PUMAs may split, merge, disappear, or newly emerge. This poses a challenge for maintaining a consistent geographic unit when a sample spans two or more PUMA vintages. For example, my sample straddles two PUMA vintages: the 2012–2021 ACS uses 2010 PUMA definitions and the 2022–2024 ACS uses 2020 PUMA definitions. No widely used crosswalk currently exists for mapping PUMA 2020 to either CZ vintage.

Motivated by the absence of a widely used PUMA 2020 crosswalk, I construct mappings for both CZ vintages covering PUMA vintages 1990 through 2020. For CZ 1990, I first replicate the Autor et al. (2019); Autor and Dorn (2013) crosswalks for PUMAs 1990, 2000, and 2010 to verify that my pipeline reproduces their published files, then extend to PUMA 2020 by chaining through PUMA 2010 using the IPUMS PUMA 2010-to-PUMA 2020 equivalency file. For CZ 2020, which is built on 2020-era county geographies, I apply the same county-as-bridge logic: I first construct a direct PUMA 2020-to-CZ 2020 crosswalk through the 2020 county, then extend to PUMA 2010 by chaining through PUMA 2020.

Two ideas unify the construction of the crosswalks. First, the allocation factor in each crosswalk is a population share: for a (PUMA, CZ) pair, it equals the share of the PUMA's population residing in that CZ, measured from the decennial census that defined the PUMA vintage. For example, a PUMA 2020 that splits 70/30 across two CZ 2020s generates two rows in the crosswalk with factors 0.70 and 0.30. The allocation factor therefore has a clean probabilistic interpretation: it is the probability that a person drawn at random from the PUMA lives in that CZ. Second, when the PUMA vintage and the CZ vintage share

a common county base, I merge directly through the county; when they do not, I chain through an intermediate PUMA vintage using the IPUMS PUMA equivalency files. In the latter case, the allocation factor for a (source PUMA, CZ) pair is the product of the source-PUMA-to-intermediate-PUMA population share and the intermediate-PUMA-to-CZ population share, summed over all intermediate PUMAs that the source PUMA overlaps. This sum is the share of the source PUMA's population residing in that CZ.

The subsections document the construction and usage of each crosswalk.

Appendix E.2 PUMA 1990 → CZ 1990

Approach. I construct a crosswalk from PUMA 1990 to CZ 1990 using the 1990 county as the bridge and 1990 census population shares as allocation factors.

Inputs.

- `geocorr1990.csv`: Geocorr 1990 PUMA 1990-to-county 1990 allocation file, weighted by 1990 census population; source: Missouri Census Data Center, Geocorr 1990 (`https://mcdc.missouri.edu/applications/geocorr1990.html`).
- `counties90_cz90.csv`: county 1990-to-CZ 1990 mapping using the ERS90 classification (Tolbert and Sizer, 1996); obtained from the Penn State Commuting Zones repository (Fowler and Jensen, 2020), `https://sites.psu.edu/psucz/data/`.

Steps.

1. Load the Geocorr 1990 file. Each row is a (PUMA 1990, county 1990) intersection with an allocation factor equal to the share of the PUMA 1990's population residing in that county 1990.
2. Merge the county 1990-to-CZ 1990 mapping on the five-digit county FIPS code.
3. Collapse to the (PUMA 1990, CZ 1990) level by summing the PUMA 1990-to-county 1990 allocation factors over all counties 1990 belonging to the same CZ 1990. The resulting sum is the PUMA 1990-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 1990's population residing in that CZ 1990.
4. Verify that the allocation factors sum to one within each PUMA 1990.

Output and Usage. The crosswalk is saved as `puma1990_cz1990_crosswalk.csv` and `puma1990_cz1990_crosswalk.dta`, with columns `puma1990`, `cz1990`, and `afactor`. The `puma1990` column is a six-character string formed by concatenating the two-digit state FIPS code with the four-digit 1990 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The crosswalk matches Autor and Dorn (2013)'s published `cw_puma1990_czone.dta` exactly: 100 percent of PUMA–CZ pairs appear in both files, with a mean absolute difference of 0.00009 in allocation factors and a maximum of 0.002 driven by rounding. This replication confirms that the ERS90 codes used here are identical to Autor and Dorn's `czone` codes.

Description. I obtain PUMA 1990-to-county 1990 population shares from the Missouri Census Data Center's Geocorr tool, which queries the MABLE (Master Area Block Level Equivalency) database. I select PUMA A (5% Sample) as the source geography, county as the target geography, and 1990 census population as the weighting variable. For every (PUMA 1990, county 1990) intersection, the extract reports an allocation factor equal to the share of the PUMA 1990's population residing in that county 1990, computed directly from block-level populations within MABLE. By construction, these shares sum to one within each PUMA 1990.

The county-to-CZ mapping uses the ERS90 classification, which is the 1990 commuting zone delineation originally constructed by Tolbert and Sizer (1996) at the USDA Economic Research Service through hierarchical cluster analysis of county-to-county commuter flows from the 1990 Census journey-to-work data. I obtain the file from the Penn State Commuting Zones repository assembled by Fowler and Jensen (2020), which distributes the ERS90 delineation together with delineations from other agencies and with fit statistics for each commuting zone. Because counties 1990 nest exactly within CZs 1990, assigning each (PUMA 1990, county 1990) row to a CZ 1990 is a simple merge on the five-digit county FIPS code. I then collapse to the (PUMA 1990, CZ 1990) level by summing the PUMA 1990-to-county 1990 allocation factors over all counties 1990 sharing the same CZ 1990. The resulting `afactor` is the share of the PUMA 1990's population residing in that CZ 1990, interpretable as the probability that a person drawn at random from the PUMA 1990 lives in that CZ 1990. This construction involves no further approximation beyond the block-level weighting already contained in the Geocorr input. PUMAs 1990 do not nest within counties 1990, which is why the Geocorr step is needed in the first place: a single PUMA 1990 can span multiple counties 1990, and those counties 1990 may in turn belong to different CZs 1990.

Appendix E.3 PUMA 2000 → CZ 1990

Approach. I construct a crosswalk from PUMA 2000 to CZ 1990 using the 2000 county as the bridge and 2000 census population shares as allocation factors. Because the CZ 1990 file uses 1990-vintage county FIPS codes, I first reconcile the small number of county code changes that occurred between 1990 and 2000 before merging.

Inputs.

- `geocorr2000.csv`: Geocorr 2000 PUMA 2000-to-county 2000 allocation file, weighted by 2000 census population; source: Missouri Census Data Center, Geocorr 2000 (https://mcdc.missouri.edu/applications/geocorr2000.html).
- `counties90_cz90.csv`: county 1990-to-CZ 1990 mapping using the ERS90 classification (Tolbert and Sizer, 1996); obtained from the Penn State Commuting Zones repository (Fowler and Jensen, 2020), https://sites.psu.edu/psucz/data/.

Steps.

1. Load the Geocorr 2000 file. Each row is a (PUMA 2000, county 2000) intersection with an allocation factor equal to the share of the PUMA 2000's population residing in that county 2000.
2. Identify county 2000 FIPS codes that do not appear in the county 1990-to-CZ 1990 mapping and remap them to their 1990 equivalents: Denali Borough, AK (02068); Skagway-Hoonah-Angoon, AK (02232); Yakutat City and Borough, AK (02282); and Miami-Dade County, FL (12086). These four remappings are documented in the Census Bureau's county code change files (https://www.census.gov/programs-surveys/geography/technical-documentation/county-changes.January_2020.html).
3. Merge the county 1990-to-CZ 1990 mapping on the reconciled five-digit county FIPS code.
4. Collapse to the (PUMA 2000, CZ 1990) level by summing the PUMA 2000-to-county 2000 allocation factors over all counties 2000 belonging to the same CZ 1990. The resulting sum is the PUMA 2000-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 2000's population residing in that CZ 1990.
5. Verify that the allocation factors sum to one within each PUMA 2000.

Output and Usage. The crosswalk is saved as `puma2000_cz1990_crosswalk.csv` and `puma2000_cz1990_crosswalk.dta`, with columns `puma2000`, `cz1990`, and `afactor`. The `puma2000` column is a six-character string formed by concatenating the two-digit state FIPS code with the four-digit 2000 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The crosswalk matches Autor et al. (2019)'s published `cw_puma2000_czone.dta` exactly: 2,880 PUMA–CZ pairs appear in both files with no pairs unique to either, and the mean absolute difference in allocation factors is 0.000087 (maximum 0.002, driven by rounding).

Description. I obtain PUMA 2000-to-county 2000 population shares from the Missouri Census Data Center's Geocorr 2000 tool, which queries the MABLE (Master Area Block Level Equivalency) database. I select PUMA (5% Sample) as the source geography, county as the target geography, and 2000 census population as the weighting variable. For every (PUMA 2000, county 2000) intersection, the extract reports an allocation factor equal to the share of the PUMA 2000's population residing in that county 2000, computed directly from block-level populations within MABLE. By construction, these shares sum to one within each PUMA 2000.

Merging the Geocorr 2000 file with the county 1990-to-CZ 1990 mapping is not a direct join on county FIPS because county boundaries and FIPS codes changed slightly between 1990 and 2000. Four counties appear in the Geocorr 2000 file under codes that are not present in the 1990-vintage CZ file. Denali Borough, AK (02068) was carved out of Yukon-Koyukuk (02290) in 1990; I remap 02068 to 02290 so that Denali's 2000 population is assigned to the CZ 1990 containing its pre-1990 parent. Skagway-Hoonah-Angoon, AK (02232) and Yakutat City and Borough, AK (02282) were both created in 1992 when the former Skagway-Yakutat-Angoon borough (02231) was partitioned; I remap both 02232 and 02282 to 02231. Miami-Dade County, FL (12086) was renamed from Dade County (12025) in 1997 with no change in territory; I remap 12086 to 12025. Two additional 1990-vintage codes appear in the CZ file but not in the Geocorr 2000 file: Yellowstone National Park, MT (30113), which was dissolved in 1997 and absorbed into Gallatin and Park counties, and South Boston City, VA (51780), which reverted to Halifax County in 1995. These simply drop out of the merge because no PUMA 2000 row references them. The remapping is documented in the Census Bureau's county code change files.

After the merge, I collapse to the (PUMA 2000, CZ 1990) level by summing the PUMA 2000-to-county 2000 allocation factors over all counties 2000 sharing the same CZ 1990. The

resulting `afactor` is the share of the PUMA 2000's population residing in that CZ 1990, interpretable as the probability that a person drawn at random from the PUMA 2000 lives in that CZ 1990. This construction involves no further approximation beyond the block-level weighting already contained in the Geocorr input. The final `afactor` is rounded to three decimal places.

Appendix E.4 PUMA 2010 → CZ 1990

Approach. I construct a crosswalk from PUMA 2010 to CZ 1990 by chaining through PUMA 2000, using the PUMA 2010-to-PUMA 2000 equivalency and the PUMA 2000-to-CZ 1990 crosswalk built in Appendix Appendix E.3.

Inputs.

- `puma2k_puma2010.csv`: PUMA 2010-to-PUMA 2000 equivalency file, weighted by 2010 census population; source: Missouri Census Data Center (https://mcdc.missouri.edu/geography/PUMAs.html).
- `puma2000_cz1990_crosswalk.csv`: population-weighted PUMA 2000-to-CZ 1990 crosswalk constructed in Appendix Appendix E.3.

Steps.

1. Load the PUMA 2010-to-PUMA 2000 equivalency file. Each row is a (PUMA 2010, PUMA 2000) pair with an allocation factor (`AFACT2`) equal to the share of the PUMA 2010's 2010 census population residing in that PUMA 2000.
2. Load the PUMA 2000-to-CZ 1990 crosswalk, which reports for each (PUMA 2000, CZ 1990) pair the share of the PUMA 2000's population residing in that CZ 1990.
3. Merge the two crosswalks on the six-character `puma2000` code.
4. Compute the combined allocation factor as the product of the PUMA 2010-to-PUMA 2000 factor and the PUMA 2000-to-CZ 1990 factor.
5. Collapse to the (PUMA 2010, CZ 1990) level by summing the combined allocation factors over all PUMA 2000 intermediaries sharing the same CZ 1990. The resulting sum is the PUMA 2010-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 2010's population residing in that CZ 1990.
6. Verify that the allocation factors sum to one within each PUMA 2010.

Output and Usage. The crosswalk is saved as `puma2010_cz1990_crosswalk.csv` and `puma2010_cz1990_crosswalk.dta`, with columns `puma2010`, `cz1990`, and `afactor`. The `puma2010` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2010 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The crosswalk matches Autor et al. (2019)'s published `cw_puma2010_czone.dta` closely: 4,077 PUMA–CZ pairs appear in both files with no pairs unique to either, and the mean absolute difference in allocation factors is 0.000178 (maximum 0.001564). The small residual reflects compounded rounding across the two-step allocation and is consistent with the three-decimal precision applied at each step.

Description. PUMA 2010 boundaries are aligned with the 2010 decennial census geography, while CZ 1990 boundaries are aligned with the 1990 decennial census geography. Rather than map PUMA 2010 directly to 1990-vintage counties, I chain through PUMA 2000, for which I have already built a crosswalk to CZ 1990 in Appendix Appendix E.3, and I rely on the Missouri Census Data Center's PUMA 2010-to-PUMA 2000 equivalency file to bridge the first link.

The PUMA 2010-to-PUMA 2000 equivalency file is published by the Missouri Census Data Center. For each (PUMA 2010, PUMA 2000) pair, it reports the share of the PUMA 2010's 2010 census population residing in that PUMA 2000, computed directly from census block-level populations within the MABLE (Master Area Block Level Equivalency) database. By construction, these shares sum to one within each PUMA 2010. I then merge this file with the PUMA 2000-to-CZ 1990 crosswalk from Appendix Appendix E.3, which is itself built from block-level populations.

The combined allocation factor for a (PUMA 2010, PUMA 2000, CZ 1990) triple is the product of the PUMA 2010-to-PUMA 2000 share and the PUMA 2000-to-CZ 1990 share. Summing over PUMA 2000 intermediaries within a common CZ 1990 yields the share of the PUMA 2010's population residing in that CZ 1990, which is the `afactor`. The final `afactor` is rounded to three decimal places.

Appendix E.5 PUMA 2020 → CZ 1990

Approach. I construct a crosswalk from PUMA 2020 to CZ 1990 by chaining through PUMA 2010, using the PUMA 2010-to-PUMA 2020 equivalency from IPUMS and the PUMA 2010-to-CZ 1990 crosswalk built in Appendix Appendix E.4.

Inputs.

- `PUMA2010_PUMA2020_crosswalk.xls`: IPUMS PUMA 2010-to-PUMA 2020 equivalency file; source: IPUMS USA, PUMA composition files (https://usa.ipums.org/usa/volii/pumas20.shtml).
- `puma2010_cz1990_crosswalk.csv`: population-weighted PUMA 2010-to-CZ 1990 crosswalk constructed in Appendix Appendix E.4.

Steps.

1. Load the IPUMS PUMA 2010-to-PUMA 2020 equivalency file. Each row is a (PUMA 2010, PUMA 2020) pair with a variable `pPUMA20_Pop20` equal to the percentage of the PUMA 2020's 2020 population residing in that PUMA 2010. Divide by 100 to obtain the allocation factor.
2. Exclude Puerto Rico PUMAs, which are not covered by CZ 1990.
3. Load the PUMA 2010-to-CZ 1990 crosswalk, which reports for each (PUMA 2010, CZ 1990) pair the share of the PUMA 2010's population residing in that CZ 1990.
4. Merge the two crosswalks on the seven-character `puma2010` code.
5. Compute the combined allocation factor as the product of the PUMA 2020-to-PUMA 2010 factor and the PUMA 2010-to-CZ 1990 factor.
6. Collapse to the (PUMA 2020, CZ 1990) level by summing the combined allocation factors over all PUMA 2010 intermediaries sharing the same CZ 1990. The resulting sum is the PUMA 2020-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 2020's population residing in that CZ 1990.
7. Verify that the allocation factors sum to one within each PUMA 2020.

Output and Usage. The crosswalk is saved as `puma2020_cz1990_crosswalk.csv` and `puma2020_cz1990_crosswalk.dta`, with columns `puma2020`, `cz1990`, and `afactor`. The `puma2020` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2020 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. I report the internal consistency check: the allocation factors sum to one within each of the 2,462 mainland PUMAs 2020 (mean 0.999998, minimum 0.998, maximum 1.004). The residual spread reflects compounded rounding across the two chained allocation steps.

Description. PUMA 2020 boundaries are aligned with the 2020 decennial census geography, while CZ 1990 boundaries are aligned with the 1990 decennial census geography. Rather than map PUMA 2020 directly to 1990-vintage counties, I chain through PUMA 2010, for which I have already built a crosswalk to CZ 1990 in Appendix Appendix E.4. The IPUMS equivalency file bridges the first link. The effective full chain is PUMA 2020 → PUMA 2010 → PUMA 2000 → CZ 1990, with no cross-vintage county remapping at any step.

The IPUMS equivalency file reports, for each (PUMA 2010, PUMA 2020) pair, the share of the PUMA 2020's 2020 population residing in that PUMA 2010, which sums to one within each PUMA 2020 by construction. I then merge this file with the PUMA 2010-to-CZ 1990 crosswalk from Appendix Appendix E.4. The combined allocation factor for a (PUMA 2020, PUMA 2010, CZ 1990) triple is the product of the PUMA 2020-to-PUMA 2010 share and the PUMA 2010-to-CZ 1990 share. Summing over PUMA 2010 intermediaries within a common CZ 1990 yields the share of the PUMA 2020's population residing in that CZ 1990, which is the `afactor`. The final `afactor` is rounded to three decimal places.

Appendix E.6 PUMA 2020 → CZ 2020

Approach. I construct a crosswalk from PUMA 2020 to CZ 2020, using the 2020 county as the bridge.

Inputs.

- `geocorr2022_2601502573.csv`: Geocorr 2022 PUMA 2020-to-county 2020 allocation file, weighted by 2020 census population; source: Missouri Census Data Center, Geocorr 2022 (https://mcdc.missouri.edu/applications/geocorr2022.html).
- `commuting-zones-2020.csv`: county 2020-to-CZ 2020 mapping. The file places the 3,222 counties and county-equivalents in the United States and Puerto Rico into 598 distinct labor markets, following the Fowler (2024) methodology with an added contiguity constraint; source: USDA Economic Research Service (https://www.ers.usda.gov/data-products/commuting-zones-and-labor-market-areas).

Steps.

1. Load the Geocorr 2022 file and drop rows with missing PUMA codes. Each remaining row is a (PUMA 2020, county 2020) intersection with an allocation factor equal to the share of the PUMA 2020's population residing in that county 2020.

2. Standardize county FIPS codes to five-digit strings in both files.

3. Merge the county 2020-to-CZ 2020 mapping onto the Geocorr file on the five-digit county FIPS code.

4. Collapse to the (PUMA 2020, CZ 2020) level by summing the PUMA 2020-to-county 2020 allocation factors over all counties 2020 belonging to the same CZ 2020. The resulting sum is the PUMA 2020-to-CZ 2020 allocation factor (`afactor`), which represents the share of the PUMA 2020's population residing in that CZ 2020.

5. Verify that the allocation factors sum to one within each PUMA 2020.

Output and Usage. The crosswalk is saved as `puma2020_cz2020_crosswalk.csv` and `puma2020_cz2020_crosswalk.dta`, with columns `puma2020`, `cz2020`, and `afactor`. The `puma2020` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2020 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The allocation factors sum to one within each of the 2,486 PUMAs 2020 (mean 0.999996, minimum 0.998, maximum 1.002). The residual spread reflects rounding in the collapse to three decimal places.

Description. PUMA 2020 boundaries are aligned with 2020 decennial census geography, and CZ 2020 is defined over 2020 counties, which include Connecticut planning regions and Puerto Rico municipios as county-equivalents. The 2020 county therefore serves as a natural bridge, and no county-code harmonization is required.

The Geocorr 2022 file reports, for each (PUMA 2020, county 2020) pair, the share of the PUMA's 2020 population residing in that county, computed from 2020 census block populations. These shares sum to one within each PUMA by construction. Because every county nests entirely within a single CZ under the ERS delineation, summing the PUMA-to-county allocation factor over all counties within a CZ yields the share of the PUMA 2020's population residing in that CZ 2020, which is the `afactor`.

Appendix E.7 PUMA 2010 → CZ 2020

Approach. I construct a crosswalk from PUMA 2010 to CZ 2020 by chaining through PUMA 2020, using the IPUMS PUMA 2010-to-PUMA 2020 equivalency and the PUMA 2020-to-CZ 2020 crosswalk built in Appendix Appendix E.6.

Inputs.

- `PUMA2010_PUMA2020_crosswalk.xls`: IPUMS PUMA 2010-to-PUMA 2020 equivalency file; source: IPUMS USA, PUMA composition files (https://usa.ipums.org/usa/volii/pumas20.shtml).
- `puma2020_cz2020_crosswalk.csv`: PUMA 2020-to-CZ 2020 crosswalk constructed in Appendix Appendix E.6.

Steps.

1. Load the IPUMS PUMA 2010-to-PUMA 2020 equivalency file. Each row is a (PUMA 2010, PUMA 2020) pair with a variable `pPUMA10_Pop20` equal to the percentage of the PUMA 2010's 2020 population residing in that PUMA 2020. Divide by 100 to obtain the allocation factor.
2. Load the PUMA 2020-to-CZ 2020 crosswalk, which reports for each (PUMA 2020, CZ 2020) pair the share of the PUMA 2020's population residing in that CZ 2020.
3. Merge the two crosswalks on the seven-character `puma2020` code.
4. Compute the combined allocation factor as the product of the PUMA 2010-to-PUMA 2020 factor and the PUMA 2020-to-CZ 2020 factor.
5. Collapse to the (PUMA 2010, CZ 2020) level by summing the combined allocation factors over all PUMA 2020 intermediaries sharing the same CZ 2020. The resulting sum is the PUMA 2010-to-CZ 2020 allocation factor (`afactor`), which represents the share of the PUMA 2010's population residing in that CZ 2020.
6. Verify that the allocation factors sum to one within each PUMA 2010.

Output and Usage. The crosswalk is saved as `puma2010_cz2020_crosswalk.csv` and `puma2010_cz2020_crosswalk.dta`, with columns `puma2010`, `cz2020`, and `afactor`. The `puma2010` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2010 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The allocation factors sum to one within each of the 2,378 PUMAs 2010 (mean 0.999993, minimum 0.998, maximum 1.002). The residual spread reflects compounded rounding across the two chained allocation steps.

Description. PUMA 2010 boundaries are aligned with the 2010 decennial census geography, while CZ 2020 is defined over 2020 counties. Rather than map PUMA 2010 directly to 2020-vintage counties, I chain through PUMA 2020, for which I have already built a crosswalk to CZ 2020 in Appendix Appendix E.6. The IPUMS equivalency file bridges the first link.

The IPUMS equivalency file reports, for each (PUMA 2010, PUMA 2020) pair, the share of the PUMA 2010's 2020 population residing in that PUMA 2020, which sums to one within each PUMA 2010 by construction. I then merge this file with the PUMA 2020-to-CZ 2020 crosswalk from Appendix Appendix E.6. The combined allocation factor for a (PUMA 2010, PUMA 2020, CZ 2020) triple is the product of the PUMA 2010-to-PUMA 2020 share and the PUMA 2020-to-CZ 2020 share. Summing over PUMA 2020 intermediaries within a common CZ 2020 yields the share of the PUMA 2010's population residing in that CZ 2020, which is the `afactor`. The final `afactor` is rounded to three decimal places.